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Early excitability changes in a novel acute model of transient focal ischemia and reperfusion in the *in vitro* isolated guinea pig brain

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Abstract

The study of the early events that characterize cerebral ischemia is limited in available experimental models. The study of neurophysiological network changes that occur in brain tissue during the early minutes that follow focal ischemia induction is restricted in the *in vivo* condition. Very simplified systems, such as *in vitro* brain slices and in isolated neurons, have been utilized for this type of studies. We describe here a new model of transient focal ischemia and reperfusion developed in the isolated guinea pig brain, maintained *in vitro* by arterial perfusion with a complex saline solution without blood cells. In this preparation, that combines the advantage of an *in vitro* preparation with the functional preservation of both vascular and neuronal compartments, the arteries of the Willis circle are directly accessible by visual control. To induce transitory focal ischemia, one medial cerebral artery (MCA) was transiently tied for 30 min, while brain activity was recorded with multiple electrodes positioned in brain areas within and outside MCA territory. Anoxic depression in ischemic areas propagated to the surrounding tissue and was associated with the abolition of evoked responses due to both functional impairment of afferent olfactory input and tissue depression. Recovery of evoked responses was obtained after MCA reperfusion. The spatial distribution of hypoxic depressions was characterized and was correlated with the extension of brain damage, defined by immunohistochemical analysis with antibodies against microtubule-associated protein (MAP-2).

We propose that the present model can be utilized to analyze brain activity changes that occur in early stages of focal brain ischemia and reperfusion.
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Keywords: Ischemia–reperfusion; Isolated guinea pig brain; Olfactory cortex; Hypoxic depression; Transitory focal ischemia

Introduction

The early phases of focal brain ischemia are crucial for the development of the following tissue damage and are mediated by mechanisms that are potential targets for therapeutic interventions. The events that occur within minutes/hours after ischemic injury are still poorly understood because of the restrictions of the available experimental models. Biochemical and molecular alterations have been investigated on simplified *in vitro* preparations incubated in deoxygenated, glucose-free medium, such as primary neurons cultures (Goldberg and Choi, 1993), co-cultures (Nagy et al., 2005), slices (Dong et al., 1988) and other *in vitro* blood–brain barrier models, like Duport's co-cultures of endothelial cells and organotypic brain slices (Duport et al., 1998) or the dynamic blood–brain barrier (Stanness et al., 1997; Stanness et al., 1999). Even if the mentioned models allow to study separately the specific mechanisms that characterized an ischemic process, in none of the above the complex interactions between extracellular tissue environment and the vascular compartment, and the occurring compensatory mechanisms are respected. *In vivo* models are closer to the real conditions even with limits well illustrated in a review by Hossmann (1998). The processes that occur in the first hours are particularly difficult to study in *in vivo* conditions, because of the influence of surgery, anesthesia and because of the laboriousness of the direct continuous control of different parameters. In few *in vivo* studies monitoring of early phases of acute focal cerebral

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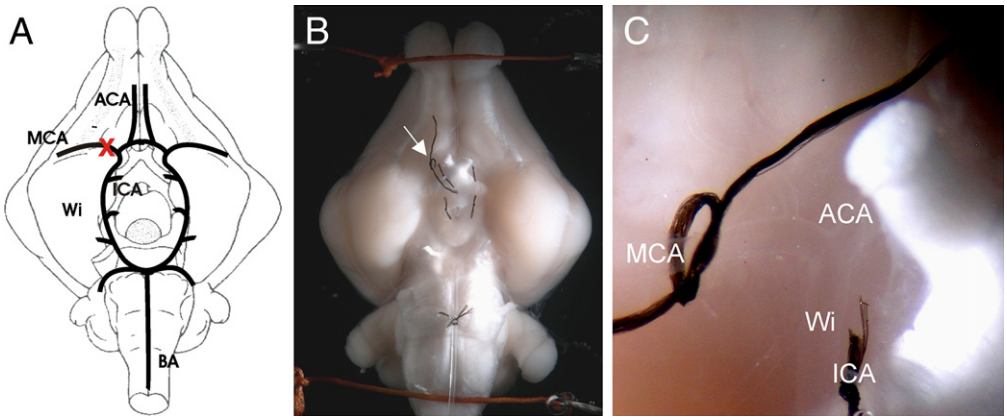

Fig. 1. (A) Schematic drawing of the arterial system of the guinea pig brain, characterized by a direct visual access to the ventral brain surface. The X marks the occluded MCA. ACA: anterior cerebral artery; MCA: medial cerebral artery; ICA: internal carotid artery; Wi: Willis circle; BA: basilar artery. (B) Stereomicroscope photograph of the isolated guinea pig brain in the incubation chamber. Both ICAs and hypophysary arteries are closed with silk threads. The basilar artery is cannulated. A loose node that will be utilized to induce ischemia is shown around the proximal tract of the MCA (arrow), and is illustrated in the enlargement in panel C.

ischemia has been approached with the use of diffusion-weighted magnetic resonance, combined with the continuous monitoring of electroencephalogram (Gyngell et al., 1995; Kohno et al., 1995).

In line with the requirements described by Hossmann (1998) for an ideal model of ischemia, we developed a model of transient focal ischemia in the isolated guinea pig brain maintained *in vitro* by arterial perfusion. In this preparation, currently used for complex electrophysiological investigations, the preservation of anatomo-morphological features of brain parenchyma has been recently assessed. The neuronal (de Curtis et al., 1991; Muhlethaler et al., 1993; de Curtis et al., 1994; van der Linden et al., 1999; Biella et al., 2002) and vascular compartments (de Curtis et al., 1998a; Di Gennaro et al., 2004; Librizzi et al., 2000), as well as the blood–brain barrier (Librizzi et al., 2001; Mazzetti et al., 2004), are structurally and functionally preserved in this preparation; the issue of the effects of brain isolation have been extensively discussed in the mentioned manuscripts. We demonstrate here that the whole brain maintained *in vitro* can be utilized to monitor the development of the ischemic process by analyzing dynamic (evoked potentials and direct-current recordings) as static (immunohistochemistry) changes induced by the proximal occlusion of the medial cerebral artery (MCA).

Materials and methods

Experiments were performed on young adult guinea pigs (Hartley strain, 200–300 g). Following anaesthesia with intraperitoneal injection of Penthotal (Pharmacia; 80 mg/kg), a transcardiac perfusion was performed with a cold oxygenated complex saline solution (see below; pH 7.1). The brain was isolated *in vitro* following the standard technique described elsewhere (Llinás et al., 1981; de Curtis et al., 1998a; Muhlethaler et al., 1993). Experiments were performed on 14 isolated guinea pig brains maintained viable for 5 h *in vitro*.

Briefly, the scalp and the bones of the skull were removed. The dorsal dura was cut and the brain was carefully extracted with the basilar artery and Willis circle *in situ*. Then the brain was positioned in an incubation chamber maintained at 15°C with its ventral surface exposed. Under stereomicroscope control, the basilar artery was cannulated and the perfusion of the brain through the resident basilar–Willy circle system was re-established, after closing with silk threads carotids and hypophysary arteries. The flow rate of the complex saline solution (126 mM NaCl, 3 mM KCl, 1.2 mM KH_2PO_4, 1.3 mM $MgSO_4$, 2.4 mM $CaCl_2$, 26 mM $NaHCO_3$, 15 mM glucose and 3% dextran MW = 70,000, oxygenated with a 95% O_2–5% CO_2

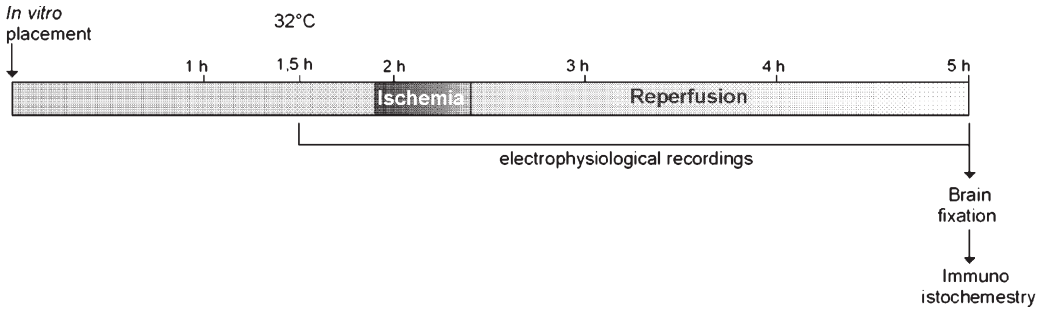

Fig. 2. Experimental protocol. After stabilization of the brain *in vitro* at 32°C, a 30-min MCA occlusion was performed. Reperfusion was restored for up to 5 h *in vitro*. At 5 h brains were fixed by immersion in a cold 4% paraformaldehyde solution for immunohistochemical analysis. Electrophysiological recordings were performed before, during and after MCA occlusion.

gas mixture, pH 7.3) supported by a peristaltic pump at either 5.5 ml/min ($n=6$) or 7 ml/min ($n=8$). The time required for brain isolation and cannulation at low temperature did not exceed 10 min. The brain temperature was raised at a rate of 0.2°C/min to the final value of 32°C.

The proximal portion of one of the MCA was carefully isolated from the surrounding dura (Fig. 1A). A silk thread (2 cm length) was slid under the MCA and a loose node was prepared around the vessel (Fig. 1B and C). To induce the transient ischemia, the MCA was closed by pulling the thread ends by tweezers. Thirty-minute MCA occlusion was induced 2 h after *in vitro* placement (Fig. 2). After 30 min, the node was released to allow reperfusion of the isolated brain up to 5 h. The experimental protocol was reviewed and approved by the Committee on Animal Care and Use and by the Ethics Committee of the Istituto Nazionale Neurologico, in accordance with national and international guidelines on care and use of laboratory animals.

Extracellular direct-current (DC) recordings were carried out with glass micropipettes filled with 0.9% NaCl (2–3 MΩ resistance) positioned at multiple sites in the olfactory cortices of both hemispheres. Recordings were performed in the piriform cortex (PC; $n=10$), in the olfactory tubercle (OT; $n=14$) and in the lateral entorhinal cortex (EC; $n=8$) ipsilateral to MCA occlusion and in the PC contralateral to the occluded vessel ($n=10$). Different arrangements of four simultaneous recording sites were utilized in the 14 experiments, to explore different parts of the olfactory–limbic region ipsi- and contralateral to the ischemic hemisphere. Signals were amplified via a multichannel differential amplifier (Biomedical

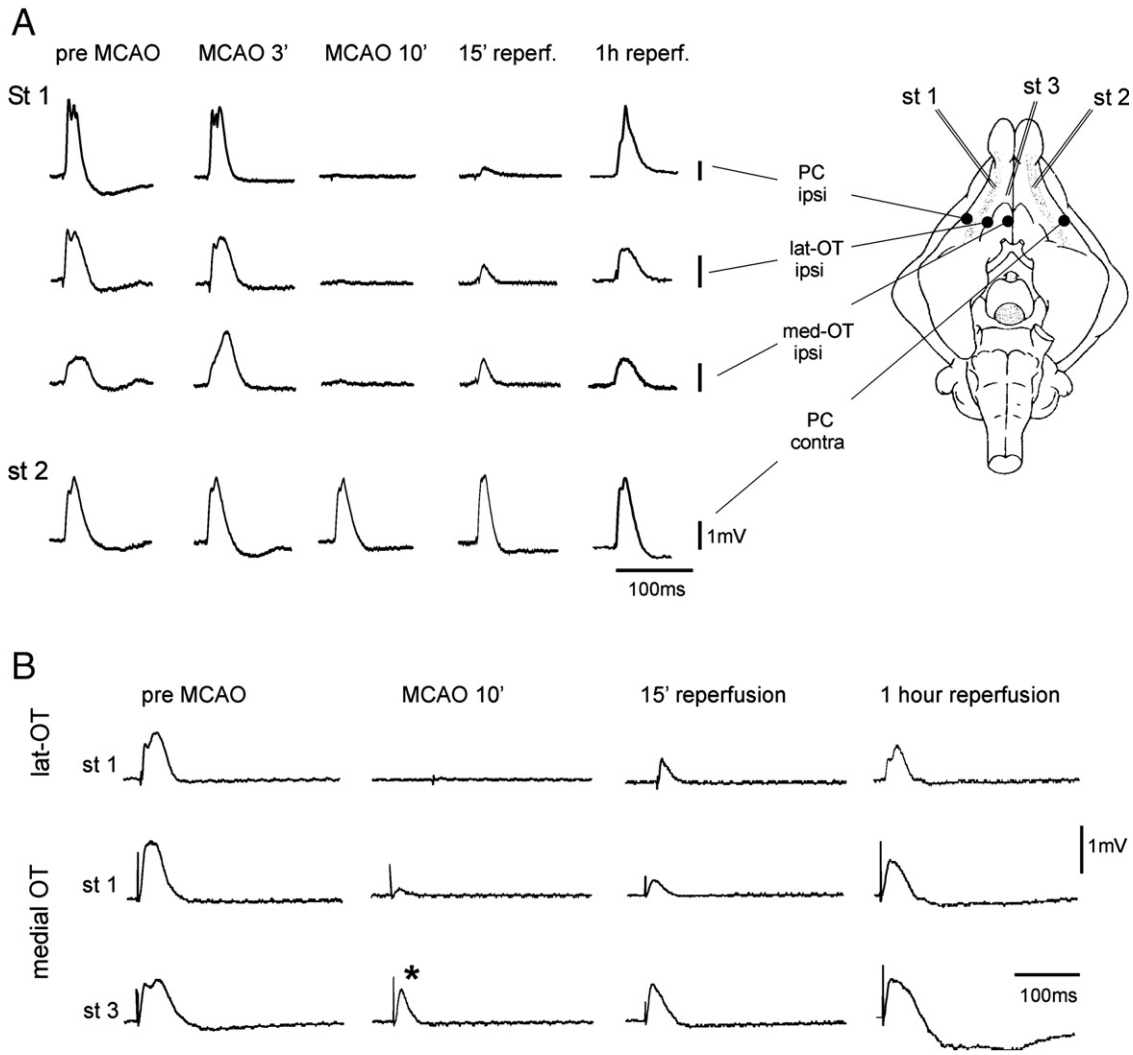

Fig. 3. Field potentials in the olfactory area during unilateral transient MCAO. (A) Evoked potentials recorded at four different sites, following stimulation of ipsilateral LOT (*st1*) and the contralateral LOT (*st2*) and local direct stimulation of the medial OT (*st3*). Electrodes were positioned in the anterior piriform cortex (PC) ipsi- and contralateral to MCA occlusion and in the ipsilateral medial and lateral olfactory tubercle (med-OT and lat-OT). Evoked responses recorded before occlusion, 3 min and 10 min after MCA occlusion and 15 min and 1 h after MCA reperfusion are illustrated. Note the preservation of the PC responses contralateral to the ischemic hemisphere. (B) Responses evoked by LOT stimulation in the lat-OT (upper traces) and in the med-OT (middle traces) are compared to the responses evoked in med-OT (lower traces) by local, direct med-OT stimulation (*st3*) close to the recording site. The polysynaptic components of the med-OT-evoked potential were abolished during MCAO and the monosynaptic potential (asterisk) was preserved. This demonstrates that the polysynaptic responses are generated in the ischemic region (possibly PC or lat-OT).

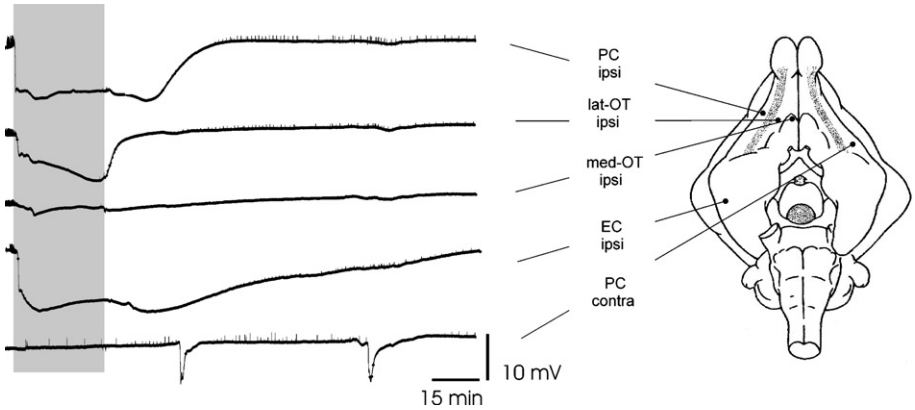

Fig. 4. Simultaneous DC recordings from four electrodes in the ipsilateral PC, med-OT, lat-OT and EC and from contralateral PC during 30-min MCA occlusion (shaded area) and subsequent reperfusion. HDs were seen in PC, lat-OT and EC ipsilateral to MCA occlusion. Fast and completely reversible SDs are shown in the contralateral PC.

Engineering, Thornwood, NY, USA) and were acquired utilizing software developed by Dr. Vadym Gnatkovsky in our laboratory (ELPHO™). Bipolar silver stimulating electrodes positioned bilaterally on each lateral olfactory tract (LOT) and in the medial OT of the ischemic hemisphere were utilized to evoke field potentials in olfactory cortices before, during and after ischemia.

At the end of the experiment (Fig. 2), the brain was fixed by immersion in a cold 4% paraformaldehyde solution in phosphate buffer (PB; 0.1 M, pH 7.4). The day after, the brain was cut in serial coronal sections (50 μm thick) with a vibratome (VT 1000S; Leica Heidelberg, Germany) throughout its rostro-caudal extension. Slices were immunoreacted for a microtubule-associated protein (MAP-2; Pettigrew et al., 1996; Bernhardt and Matus, 1984; De Camilli et al., 1984), previously utilized as an early marker of ischemic damage (Li et al., 1998; Pettigrew et al., 1996; Wiche, 1989; Kitagawa et al., 1989). Free-floating coronal sections were pre-incubated for 10 min in 3% H_2O_2 in phosphate-buffered saline (PBS) at pH 7.4, to inactivate endogenous peroxidase and then rinsed in PBS. Nonspecific sites were blocked in PBS containing 10% normal horse serum (NHS) and 0.2% Triton X-100. Sections were then incubated overnight at 4°C with anti-MAP-2 primary antibody (1:1000 MAP-2 clone AP 20, Bio-Optica, Fremont, CA, USA) diluted in 1% NHS in PBS. Then the sections were rinsed in PBS and were incubated for 75 min in monoclonal biotinylated horse anti mouse IgG (1:200; Vector Laboratories Inc., Burlingame, CA, USA). The avidin–biotin peroxidase protocol (ABC kit, Vector Labs) was applied, using 3,3′-diaminobenzidine tetrahydrochloride (DAB; Sigma, St. Louis, MO, USA) as chromogen. After staining, sections were dehydrated, cleared with xylene and coverslipped with dibutyl phthalate xylene (DPX). In order to control the specificity of the anti-MAP-2 antibody, series of sections were processed omitting the primary antibody. Specific staining was detected neither in these sections nor by replacing it with normal serum (1:100).

For immunofluorescence-staining, free-floating sections were incubated first for 1 h in PBS containing 10% NHS and 0.2 % Triton X-100, and then overnight in the primary antibody MAP-2 diluted in PBS with 1% NHS at 4°C. Sections were incubated for 2 h at room temperature in an indocarbocyanine (CY3)-conjugated horse–anti-mouse IgG solution (1:600, Jackson Immunoresearch Laboratories, West Grove, PA, USA) and, after repeated rinsing, were mounted in Fluorsave (Calbiochem, San Diego, CA, USA) and were examined under a confocal laser scanning microscope (Bio-Rad, Hemel

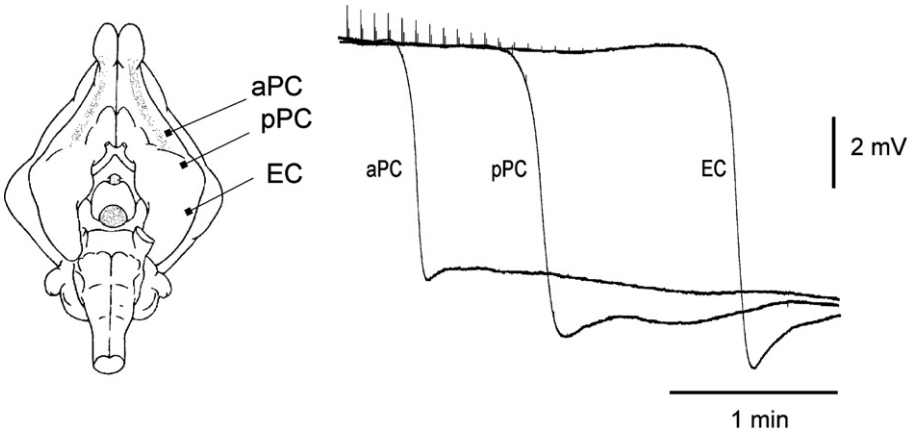

Fig. 5. Simultaneous DC recordings from three electrodes positioned in the anterior PC (aPC), in the posterior PC (pPC) and in the EC after MCAO (left panel). The timing of HD onsets in the three cortical positions demonstrates a clear rostro-caudal propagation.

Hemstead, UK) equipped with an argon/krypton gas laser mounted on a light microscope (Eclipse E600; Nikon, Tokio, Japan). CY3 was excited at 550 nm lines. Confocal image series were recorded with Bio-Rad Lasersharp 2000 software.

Results

The vitality of the isolated brains was assessed by recording potentials evoked by LOT stimulation before the induction of ischemia ($n=14$). As illustrated in Fig. 3, LOT-evoked responses were recorded in the PC ($n=10$), in the lateral (lat-OT; $n=9$), in the medial OT (med-OT; $n=8$) and in the rostral part of the EC ($n=8$) ipsilateral to the MCA occlusion and in the contralateral PC ($n=10$). During the experiment, no low pass filter was utilized, in order to allow recording of slow depolarizing events, such as spreading depressions (see Figs. 4 and 5). PC, rostral EC and lat-OT are vascularized exclusively by the MCA (Librizzi et al., 1999), while med-OT is supplied also from the anterior cerebral artery. As shown in Fig. 3A, the LOT-evoked responses gradually disappeared within 6 min in all recording electrodes of the hemisphere ipsilateral to the MCA occlusion (see also Table 1). In five experiments the LOT-evoked response transiently increased in amplitude just after MCA occlusion (not shown). To test whether the abolition of the LOT-evoked med-OT response was due to anoxia of the LOT (see Discussion), a stimulation electrode was positioned in the med-OT (*st3* in Fig. 3B) for direct local activation. During MCA occlusion, direct OT stimulation induced a monosynaptic response in med-OT (asterisk in bottom traces of Fig. 3B), whereas di- and poly-synaptic components were abolished. No responses were evoked in PC, lat-OT and EC by med-OT stimulation during the ischemic period (not shown). These findings and the persistence of the PC response evoked by contralateral LOT stimulation in the non-ischemic hemisphere demonstrated that the disappearance of evoked brain activity was regional and selective for the brain areas served by the occluded MCA.

The effect of transient MCA occlusion on the evoked response reversed with variable time course in different experiments. Field potential recovery (>50% control) occurred in 7 out of 10 experiments in PC, in 7 out of 9 experiments in lat-OT; in 4 out of 7 experiments in EC. No recovery or partial recovery (<then 50% control amplitude) were observed in 2 experiments in which edema occurred.

In all the experiments, after MCA occlusion, large and persistent negative-going voltage deflections were recorded in PC, lat-OT and EC of the hemisphere ipsilateral to MCA occlusion (Fig. 4 and Table 1). The slow depolarizations resemble previously described ischemic depressions (Umegaki et al., 2005; Van Harreveld and Stamm, 1953; Marshall, 1959; Hansen and Zeuthen, 1981) and showed a peculiar pattern of cortical spread. We will refer to them as hypoxic depression (HD; Somjen, 2004). As shown in the typical experiment illustrated in Fig. 4, HD appeared first in the PC ca. 4 min after MCA occlusion (see Table 1) as a sharp negative shift followed by a second negative deflection, with the shape of a inverted saddle with two maxima (Herreras and Somjen, 1993). HDs were observed in lat-OT and in the EC with different delays (see Table 1). HD was never observed in the contralateral PC and in the med-OT. The large depolarizations in PC and lat-OT were occasionally preceded by a brief positive deflection of less than 2 mV ($n=5$; not shown). Fig. 5 shows the slow rostro-caudal spread of HD, which starts in the anterior PC and follows in the posterior PC and the EC. The delay between the onset of HD at the three sites is compatible with a cortical spreading depression, as described by Leão.

The recovery time of HDs was variable in different experiments and in different structures. After MCA reperfusion onset (Fig. 4) recovery from HD to 75% of the control value was 88.02±8.88 min in PC (mean±SE; $n=8$), 77.17±16.95 min in lat-OT ($n=5$) and 122.6±18.80 min in EC ($n=5$). Complete recovery was obtained in 58% of the tests. The recovery of HD was paralleled by the resurgence of LOT-evoked field potentials. Partial recovery to less than 75% pre-ischemic control values or no recovery was observed in PC ($n=2$), lat-OT ($n=3$) and in EC ($n=2$). Brief and transient depolarizations were occasionally observed during reperfusion in regions remote from the ischemic areas, such as the contralateral PC ($n=4$), the lat-OT ($n=1$), the med-OT ($n=5$) and the EC ($n=1$). The time course of such events (6.33±0.61 min; $n=11$) was

Table 1
Time-course and amplitudes of hypoxic depolarization (HD) and delays of disappearance/recovery of field potential responses

	HD onset	HD duration (recovery >75%)	HD max amplitude	LOT-evoked FP abolition	LOT-evoked FP recovery onset
PC	3.75±0.75 ($n=10$)	88.02±8.88 ($n=8$)	21.7 mV±2.8 mV ($n=10$)	4.58±0.9 ($n=10$)	15.37±3.05 ($n=8$)
lat-OT	4.23±0.85 ($n=8$)	77.17±16.95 ($n=5$)	14.5 mV±2.1 mV ($n=8$)	4.20±1.0.3 ($n=8$)	13.27±5.70 ($n=8$)
EC	8.65±1.28 ($n=7$)	122.60±18.80 ($n=5$)	18.4 mV±2.6 mV ($n=7$)	4.2±1.22 ($n=6$)	25.33±4.87 ($n=4$)
med-OT	–	–	–	5.32±0.83 ($n=8$)	6.32±2.2 ($n=8$)

Values are expressed as mean±standard error (SE). HD onset was calculated from the beginning of MCA occlusion. HD duration was measured between the start and the end point of steady recovery of the trace to at least 75% of the control value. The amplitude of HDs was measured on the voltage difference between the start point and the maximum negative peak (see also Fig. 4). LOT-evoked field potential (FP) abolition was calculated from MCA occlusion onset. LOT-evoked FP recovery times were calculated from the end of MCA occlusion.

much faster than HDs, and closely resembled non-ischemic spreading depressions (Leao, 1986; Leao, 1951; Somjen, 2001).

Morphological brain tissue evaluation at the end of the electrophysiological experiments confirmed the presence of post-ischemic changes in the structures in which HD occurred. MAP-2 immunoreactivity in the unaffected control hemisphere showed homogeneous staining across cortical regions, fainter in the striatum (Pettigrew et al., 1996), whereas the MAP-2 signal was markedly reduced in brain areas supplied by the occluded MCA (Fig. 6A; see Discussion). The evaluation of the superimposed areas of MAP-2 immunoreactive reduction in the different experiments at two different rostro-caudal coronal levels confirmed that the area of injury was reproducible (Fig. 6B). In all experiments the marginal, sub-pial part of the cortex was preserved. No correlation between the extension of the MAP-2-negative area and the recovery pattern of HD was not analyzed in details.

High-power images of selected brain areas are illustrated in Fig. 7 (neocortex) and Fig. 8 (piriform cortex and neocortex). MAP-2 staining in neocortex was intense in large dendrites (but not in the somata) of superficial and deep layer neurons in the control hemisphere and in non-ischemic regions ipsilateral to the occluded MCA (Figs. 7A and C and 8A and C). In cortical areas in which HDs occurred, staining in the molecular layer was preserved, while neuropile staining was markedly reduced in mid-deep layers where dendrite MAP-2 labeling was sparse and characterized by clustered chains (beads) of immunoreactive product (Fig. 7B and D). Neurons of deep layers of the affected ischemic hemisphere typically showed decreased dendritic (Fig. 8D) and increased perikaryal MAP-2 immunostaining (Fig. 8B) compared to the control side (Fig. 8A and C). MAP-2 immunoreactivity at the soma of neurons was also evident in the transition areas around the ischemic core, such as the med-OT. In all experiments, thionine-stained sections adjacent to those processed for MAP-2 (Fig. 8C and D) did not show signs of neuronal depletion in the ischemic area (Fig. 7E and F). Indeed, no nuclear picnosis or cytoplasmic darkening, interpretable as signs of cell suffering, were evident by high magnification inspection after MCA occlusion and reperfusion.

Discussion

The present study demonstrates that reproducible transitory focal ischemia of the MCA territory can be induced in the isolated guinea pig brain maintained *in vitro* by arterial perfusion. In the isolated whole guinea pig brain the complex relations among vascular, extracellular and intracellular compartments are functionally and structurally preserved (Librizzi et al., 2000; Muhlethaler et al., 1993; de Curtis et al., 1998a; Librizzi et al., 2001; Mazzetti et al., 2004). Several studies performed in the last 15 years demonstrated that brain activities

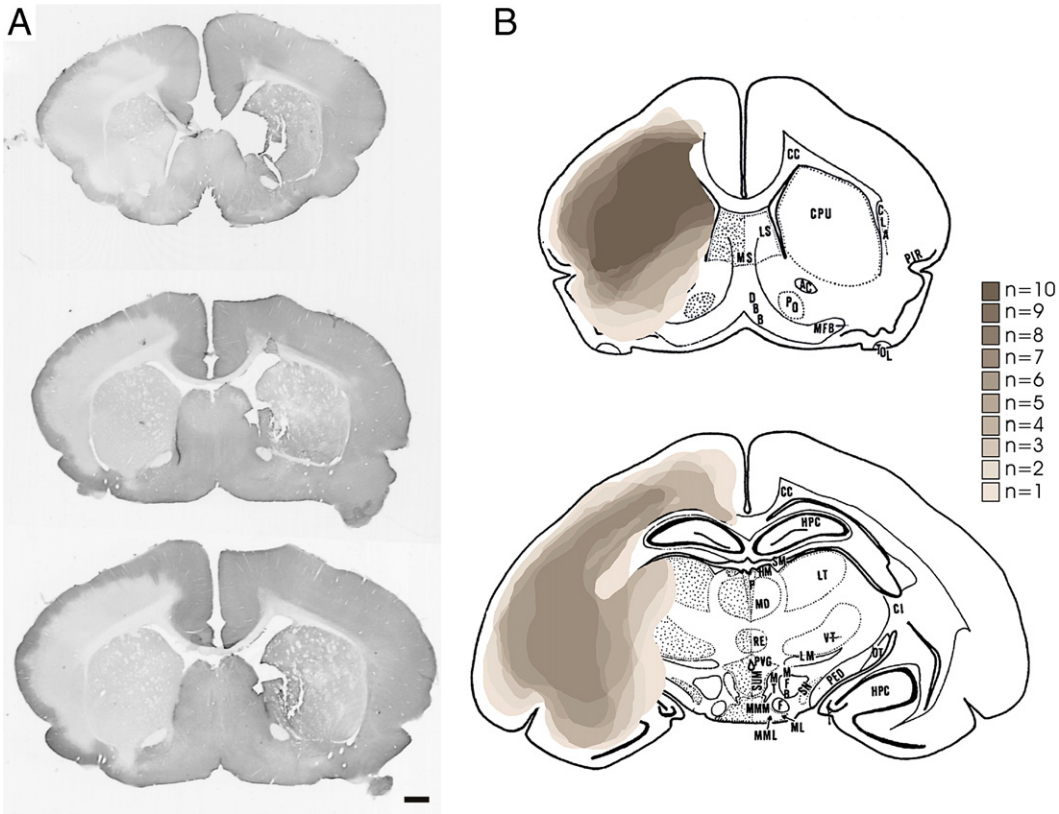

Fig. 6. (A) Low power scanner images of 50 μm coronal sections immunoreacted for MAP-2 at three different rostro-caudal levels. Scale bar = 1 mm. (B) MAP-2 negative areas of different experiments ($n = 10$ in upper panel and $n = 7$ in lower panel) superimposed on two different coronal plates of the guinea pig atlas. Different intensities of grey shading represent the MAP-2 negative area reproduced in the number of experiments indicated on the right of the calibration scale. The area colored by the darkest gray in each section is MAP-2 negative area reproduced in all the experiments.

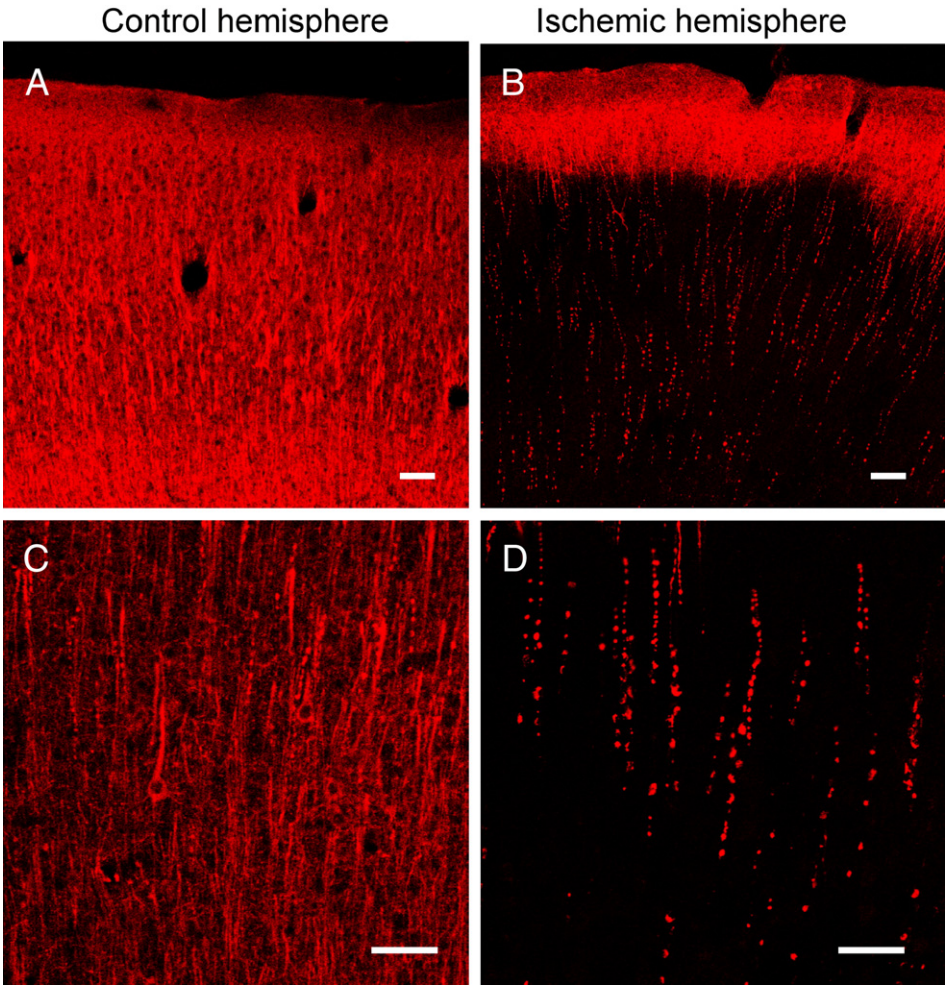

Fig. 7. Confocal fluorescence microscope images of MAP-2 immunoreacted piriform cortex (PC) sections (50 μm thick) cut from brains fixed at the end of the electrophysiological experiment. On the left, fluorescent MAP-2 images from the hemisphere contralateral to ischemic side (control hemisphere). (A) Superficial mid-layers (I–IV) of the PC. (C) Deep PC layers (control hemisphere). On the right, fluorescent MAP-2 staining of homologous areas in the ischemic hemisphere. (B) Superficial mid-layers showed progressive reduction of MAP-2 immunoreactivity (10×). (D) Deep PC layers. Neuropile staining was preserved in I–II layers, but was nearly absent in deeper layers. Dendritic staining is characterized by clustered chains. Calibration bar = 100 μm.

recorded in the olfactory and limbic regions of the isolated guinea pig brain preparation are similar to the potentials observed *in vivo* and on *in vitro* slices (see Biella et al., 2002, 2003; de Curtis et al., 1991, 1998b; van der Linden et al., 1999). In the present study, recordings were performed in these brain areas that are supplied by the MCA. Preservation of the vascular compartment (de Curtis et al., 1998a; Librizzi et al., 2000) and of the blood–brain barrier (Librizzi et al., 2001, 2006; Mazzetti et al., 2004) was also recently demonstrated. As discussed in the mentioned studies, the transient ischemia imposed during brain isolation in conditions of hypothermia does not interfere with the neurophysiological and vascular parameters analyzed so far. The *in vitro* isolated guinea pig brain represents a close to *in vivo* condition, in which electrophysiological events and morphological changes can be analyzed during the development of the ischemia process.

As for any *in vitro* experimental condition, advantages and limitation intrinsic to the use of the isolated brain preparation should be clearly recognized. Experiments were conducted at 32°C to achieve the obligatory requisite to maintain the brain in an anesthetized state. This condition should be taken in consideration as a limit of our experimental setting, since it could affect and maybe protect against the development of ischemic damage. Indeed, it has been documented that mild hypothermia may have a beneficial influence on the effects of ischemia (Yunoki et al., 2003; Sick et al., 1999). The use of anesthetic drugs to maintain the isolated brain preparation in and "unconscious" state could be also utilized in principle as an alternative to low temperature. We decided to utilize hypothermic anesthesia, since most of the anesthetic drugs exert a direct pharmacological effect on brain functions that are known to interfere with normal brain activity and with the ischemic process. In our study, the temperature of the isolated brain was steady during the recordings and the effect of unilateral MCA occlusion were evaluated by comparing the two hemispheres. Endothelial shear stress is expected to be increased in the *in vitro* isolated brain, since a higher than normal flow rate is prerequisite to maintain the brain viable. This condition does not appear to exert an activating effect on the isolated brain endothelium, since expression of inducible adhesion molecules

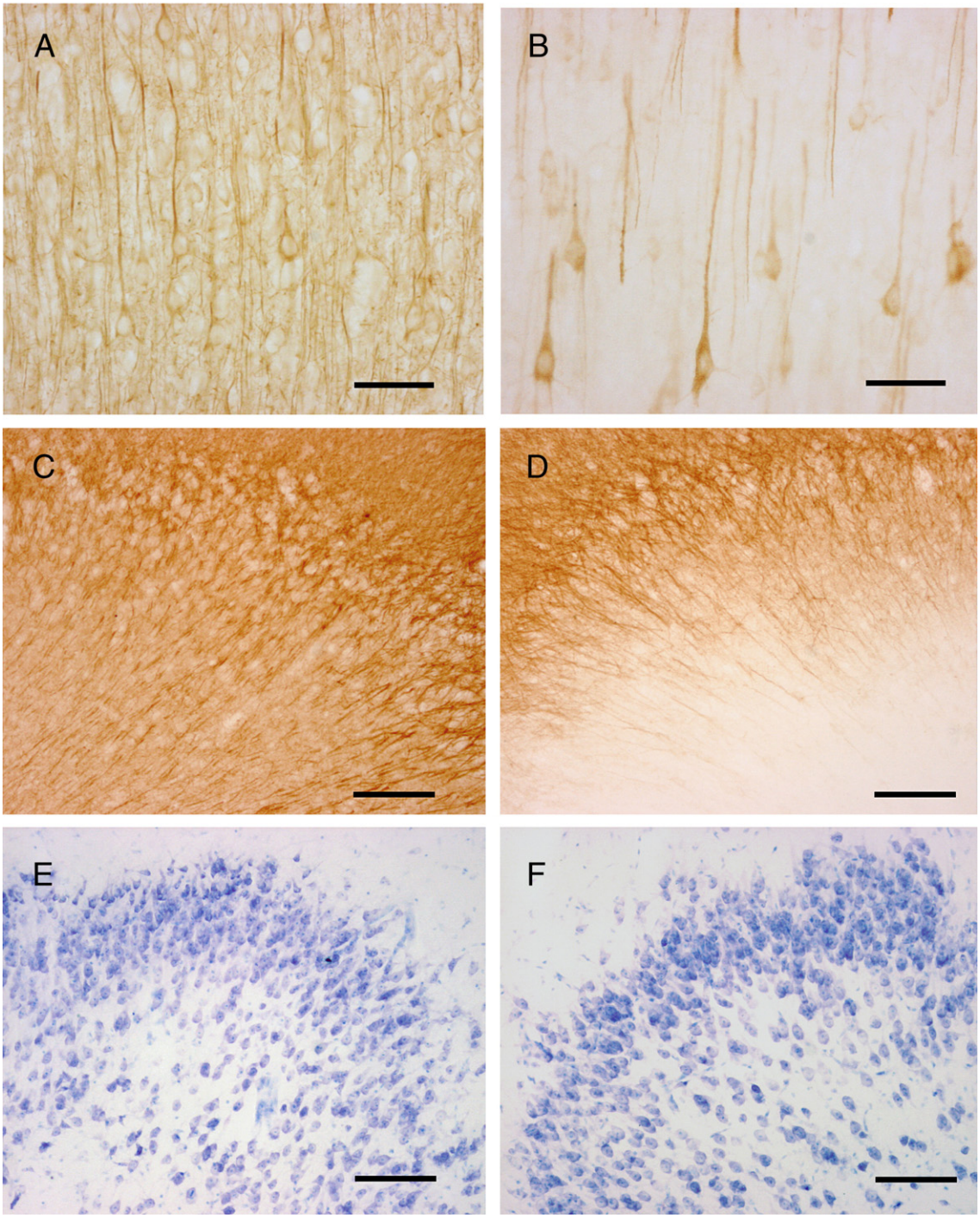

Fig. 8. Representative bright field photomicrographs (20×) of the neocortex demonstrating the loss of neuronal dendritic MAP-2 immunoreactivity and the increased of perikaryal immunostaining in the ischemic hemisphere (B) in comparison to the control contralateral side (A). In the control hemisphere, the immunoreactivity was prominent in apical dendrites but is also evident in finer dendritic branches. Photomicrographs (10×) of thionine-stained piriform cortex sections of control (E) and ischemic side (F), adjacent to the MAP-2 stained sections illustrated in panels C and D, show no signs of nuclear picnosis or cytoplasmic "darkening". Calibration bar is 100 μm in panels A and B and 50 μm in panels C–F.

is not observed in unstimulated isolated brains (Librizzi et al., 2006) and blood–brain barrier functionality is preserved (Librizzi et al., 2001; Mazzetti et al., 2004). Enhanced shear stress could be responsible for a "tonic" vasodilated condition observed in our experiments (see Librizzi et al., 2000) and should be taken in account. Increased endothelial shear stress has been shown to induce nitric oxide release from endothelial cells (Pyke and Tschakovsky, 2005). According to this hypothesis nitric oxide synthase inhibitors were found to increase vascular tone in the isolated brain preparation (Librizzi et al., 2000).

Because of the position of the isolated brain in the recording chamber (see Fig. 1), the access to the Willis circle and to the

major arterial branches is particularly facilitated and the MCA occlusion can therefore be easily performed under stereomicroscope control. The duration of vessel occlusion and re-opening can be precisely monitored because of the direct access to brain vasculature. MCA in the guinea pig has a well-defined perfusion territory (Librizzi et al., 1999), which includes the rostral part of the EC, the insular cortex, the frontal–temporal and parietal neocortex, the PC, the amygdala, the lateral OT, the caudal part of caudate and putamen, the claustrum and the nucleus accumbens. Neighboring areas served by the anterior cerebral artery and by the limbic artery (Librizzi et al., 1999) include the cingulate cortex and the adjacent apical portion of frontal cortex, the medial part of the OT, the medio-caudal part of the EC and the hippocampus. The watershed regions between MCA and the anterior and limbic arteries are potential areas of ischemic penumbra.

Histological data demonstrated a reproducible area of ischemic damage. MAP-2 was confirmed to be a good marker of the dendro-somatic neuronal injury that defines and localizes the ischemic MCA territory (Librizzi et al., 1999). MAP-2 staining alterations were coherent with those described at 15 min and 6 h in *in vivo* models of permanent MCA occlusion in the rat (Pettigrew et al., 1996; Li et al., 1998; Dawson and Hallenbeck, 1996) and during focal ischemia in the gerbil (Kitagawa et al., 1989; Tomimoto and Yanagihara, 1992). MAP-2 staining showed typical features previously described in ischemic areas, characterized by accumulation in rare enlarged and beaded dendrites (Hori and Carpenter, 1994) and increased somatic staining.

Changes in electrophysiological activities, such as evoked potential and slow DC shifts can be continuously monitored during the different phases of the experiment. Prolonged slow depolarizing events that resembled the spreading depression (SD) first described by Leão (Leao, 1951, 1986) were observed in the ischemic region. Spreading depressions are all-or-none events of massive neuronal and glial depolarization generated by diverse insults that slowly propagate in the cortex (Somjen, 2001, 2004). The potential shifts observed in our experiments were prolonged and stable during MCA occlusion and should be classified as hypoxic/ischemic spreading depression (HD; Aitken et al., 1998; Umegaki et al., 2005). The delays between HD in the different cortical areas analyzed suggest the idea of a propagating event (Figs. 4 and 5). Although hypoxic spreading depression and SD mechanisms of generation and propagation are similar (Aitken et al., 1998; Jing et al., 1994; Somjen, 2001), hypoxic spreading depression recovers only if tissue oxygenation is restored within a critical period of time, while SD is a self-limiting phenomenon that requires oxidative energy (Wang et al., 2000). Moreover, synaptic failure responsible for the activity depression develops before the onset of the wave in HD, while it accompanies Leão SD (2004). In our recordings, the amplitude of the voltage changes (10–20 mV) and the "inverted saddle" or "a peak and a plateau" time course of the events were compatible with hypoxic spreading depression (Somjen, 2001; Herreras and Somjen, 1993). It is interesting to note that terminal depolarizations with no recovery were observed mostly in the PC, where HD firstly occurred and showed the largest amplitude.

Potentials evoked by LOT stimulation vanished during HD. The time required for LOT-evoked potentials to disappear in different cortical areas did not correlate with the onset of the local HD and was nearly simultaneous in all recorded areas (see Table 1). For instance, HD onset in the rostral EC followed HD onset in the PC; nevertheless, LOT-evoked potential loss in the EC preceded HD onset and coincided to the abolition of evoked potentials in the PC. The uncoupling between HD and LOT-evoked response disappearance (see also Young and Somjen, 1992) suggests that the abolition of evoked responses in different regions served by MCA is not initiated by the local HD events, but most probably depends on the presynaptic impairment of LOT fiber-mediated input (Rosen and Morris, 1993). According to this hypothesis, the recovery of the LOT-evoked response paralleled the local recovery from HD. The loss of evoked responses is also expected to be reinforced by SD (Somjen, 2004).

Some variability in MCAO-induced electrophysiological and morphological changes was observed in our experiments. This is not unexpected, since irregularity in the extent of the affected area is a common finding in ischemia studies performed *in vivo* in different animal species (Aspey et al., 2000; Tsuchiya et al., 2003; Harada et al., 2005; Nabavi et al., 2001). Differences in recovery times of both LOT-evoked potentials and HDs could be attributed to morphological inter-individual variability of (i) the size of large-capacity vessels that compensate the MCA occlusion, (ii) the extension of proximal and distal collateral circles and (iii) the differences in the site of origin of arterial branches that stem from the MCA and supply the olfactory region.

Responses to LOT-stimulation in the transition area med-OT (not supplied by MCA) were also abolished in spite of the fact that no HD was observed in this region. This phenomenon could be interpreted as due to ischemic depression of activity mediated by the fibers of the LOT, without the contribution of local HD. To verify whether med-OT was functionally preserved, direct stimulation of the med-OT was performed. Med-OT field responses were found to be preserved following local stimulation. This further confirms that the input carried by LOT fibers into med-OT is lost during ischemia, while local activity is preserved.

In six experiments, we observed brief SDs with fast recovery times in cortical areas not served by MCA. These events could represent local electrical phenomena in penumbra or remote brain areas spared by the direct ischemic insult. The present study represents the first attempt to characterize the area of ischemic penumbra in this preparation with electrophysiological means. A better definition could be obtained with a more accurate electrophysiological mapping, extracellular ion measurements (de Curtis et al., 1998b), optical imaging (Biella et al., 2003; de Curtis et al., 1999) and immunocytochemical analysis of brain tissue especially to investigate penumbra zone.

In conclusion, a new model of focal ischemia and reperfusion can reliably be reproduced in the *in vitro* isolated guinea pig brain to study the early phases of a focal ischemia.

The preservation of the structural and functional relationship between neuronal, extracellular and vascular compartments in the whole organ confers a close-to-*in vivo* condition and allows investigating directly and continuously the viability of different brain areas during ischemia. The absence of peripheral elements, such as blood cells and plasma factors to the development of ischemia is an apparent limit of the preparation. This constrain can be considered an experimental advantage because it provides the opportunity to study the specific contribution of different factors to the pathogenesis of brain disease by introducing them individually and separately in the isolated brain, as recently attempted for the study of brain inflammation (Librizzi et al., 2006).

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